

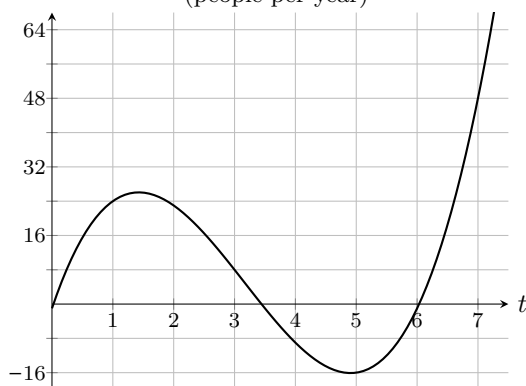
The Definite Integral

Today's Goals

- ◆ You should be able to recognize and use the notation $\int_a^b f(x) dx$ for the definite integral.
- ◆ You should be able to think about the definite integral in three different ways: as a limit of left-hand or right-hand sums, as a signed area, and as a net change of a rate function.
- ◆ You should be able to use the graphical interpretation of the definite integral to evaluate $\int_a^b f(x) dx$ when f is a function whose graph is simple or has useful symmetry.

1. The population of a tiny island country changes over time, as people immigrate and emigrate. Suppose $f(t)$ gives the rate of change of the population, where t is measured in years.

$f(t)$ = rate of change of island's population
(people per year)

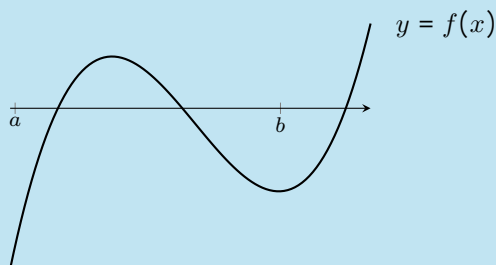


- (a) How would you visualize the exact net change in population between $t = 1$ and $t = 7$?
- (b) Use a right-hand sum with 3 rectangles to estimate the net change in population between $t = 1$ and $t = 7$.
- (c) Was the population larger at time $t = 1$ or time $t = 7$? How do you know?
- (d) Suppose the population at time $t = 2$ was 2000 people. Write an expression involving a definite integral which gives the number of people at time $t = 4$.

The Definite Integral

We have three major ways of thinking about the definite integral $\int_a^b f(x) dx$:

1. If $a < b$, we visualize it as _____.



2. If f is a rate function, then we interpret $\int_a^b f(x) dx$ as _____.

3. $\int_a^b f(x) dx$ is actually defined to be _____.

2. (a) Evaluate $\int_{-1}^2 (|t| - 1) dt$.

- (b) Suppose $|t| - 1$ gives the rate (in gallons per hour) of water flowing into and out of a tank t hours after noon on a particular day. Which of the following is a correct interpretation of $\int_{-1}^2 (|t| - 1) dt$?

- A. It is the amount of water in the tank at 2 pm.
- B. It is the amount of water in the tank at 11 am.
- C. It is (amount of water in the tank at 2 pm) – (amount of water in the tank at 11 am).
- D. It is (amount of water in the tank at 11 am) – (amount of water in the tank at 2 pm).

The graph shows a piecewise linear function $f(t)$ on the interval $[0, 20]$. The function is constant at 120 for $t \in [0, 5]$, decreases linearly from $(5, 120)$ to $(15, -180)$, and is constant at -180 for $t \in [15, 20]$.

The function $r(t)$ graphed to the left gives the rate at which water is flowing into or out of a backyard swimming pool. Time t is measured in hours, and the rate $r(t)$ is measured in gallons per hour. At time $t = 0$, there are 700 gallons of water in the pool.

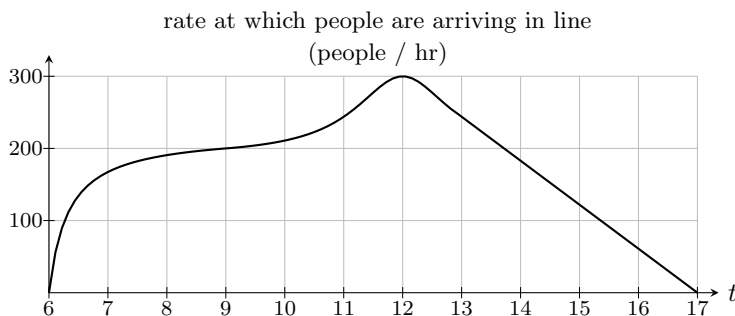
- (a) When is the amount of water in the pool greatest? At that time, how much water is in the pool?

- (b) Recall that, at $t = 0$, the pool contains 700 gallons of water. Is there ever another time T at which the pool contains 700 gallons of water? If so, what is T ? If not, why not?

- (c) How much water is in the pool at time x ? (You may leave your answer in terms of a definite integral.)

4. In the wake of the Great Depression (around 1932), many Americans lost their jobs. At that time, unemployment insurance was not in place (that began in 1936). Many people stood in lines at soup kitchens, lines so long that they wrapped around city blocks.

The graph below is a graph of $r(t)$, the rate that people are arriving to get in line at a Detroit soup kitchen. Time t is given in hours since midnight, so $t = 6$ is 6 am. (Nobody is in line at 6 am.) Soup is served between the hours of 9 am and 5 pm, and the rate at which people can be served is 200 per hour.



- (a) Approximately how long is the line when the soup kitchen opens?
- (b) Approximately how long is the line at noon?
- (c) At approximately what time is the length of the line increasing most rapidly?
- (d) At approximately what time is the line the longest?
- (e) When the line is the longest, approximately how many people are in line?
- (f) People at the end of the line routinely went home without food. How many people were left in line at closing time?
- (g) If there were enough food left to feed all people in line at 5 pm, at approximately what time would the work of the soup kitchen be done?

Express the following quantities in terms of definite integrals.

- (h) The length of the line at 8 am.
- (i) The length of the line at 10 am.
- (j) The amount of time a person arriving at 10 am must wait to be served.